CODE SPORT

This month's column continues the discussion of natural language processing.

For the past few months, we have been discussing information retrieval and natural language processing (NLP), as well as the algorithms associated with them. In this month's column, let's continue our discussion on NLP while also covering an important NLP application called 'Named Entity Recognition' (NER). As mentioned earlier, given a large number of text documents, NLP techniques are employed to extract information from the documents. One of the most common sources of textual information is newspaper articles. Let us consider a simple example wherein we are given all the newspaper articles that appeared in the last one year. The task that is assigned to us is related to the world of business. We are asked to find out all the mergers and acquisitions of businesses. We need to extract information on which companies bought over other firms as well as the companies that merged with each other. Our first rudimentary steps towards getting this information will perhaps be to look for keyword-based searches that used terms such as 'merger' or 'buys'. Once we find the sentences containing those keywords, we could then perhaps look for the names of the companies, if any occur in those sentences. Such a task requires us to identify all company names present in the document.

For a person reading the newspaper article, such a task seems simple and straightforward. Let us first try to list down the ways in which a human being would try to identify the company names that could be present in a text document. We need to use heuristics such as: (a) Company names typically would begin with capital letters; (b) They can contain words such as 'Corporation' or 'Ltd'; (c) They can be represented by letters of the alphabet separated by full stops, such as I.B.M. We could also use contextual clues such as *'X's stock price went up'* to infer that X is a business or company. Now, the question we are left with is whether it is possible to convert what constitutes our intuitive knowledge about how to look for a company's name in a text document into rules that can be automatically checked by a program. This is the task that is faced by NLP applications which try to do Named Entity Recognition (NER). The point to note is that while the simple heuristics we use to identify names of companies does work well in many cases, it is also quite possible that it misses out extracting names of companies in certain other cases. For instance, consider the possibility of the company's name being represented as IBM instead of I.B.M, or as International Business Machines. The rule-based system could potentially miss out recognising it. Similarly, consider a sentence like, "Indian Oil and Natural Gas Company decided that…" In this case, it is difficult to figure out whether there are two independent entities, namely, 'Indian Oil' and 'Natural Gas Company' being referred to in the sentence or if it is a single entity whose name is 'Indian Oil and Natural Gas Company'. It requires considerable knowledge about the business world to resolve the ambiguity. We could perhaps consult the 'World Wide Web' or Wikipedia to clear our doubts. The use of such sources of knowledge is quite common in Named Entity Recognition (NER) systems. Now let us look a bit deeper into NER systems and their uses.

Types of entities

What are the types of entities that are of interest to a NER system? Named entities are by definition, proper nouns, i.e., nouns that refer to a particular person, place, organisation, thing, date or time, such as Sandya, Star Wars, Pride and Prejudice, Cubbon Park, March, Friday, Wipro Ltd, Boy Scouts, and the Statue of Liberty. Note that a named entity can span more than one word, as in the case of 'Cubbon Park'. Each of these entities are assigned different tags such